S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026 · AI/NLP RISK ENGINE

Sentinel.

An AI risk engine that reads the world's financial news, in any language, and tells a bank what it means for its portfolio.

9 sources

news, filings, central banks and social posts, in 65 languages, no API keys

3 scores

for every company in every story: Sentiment, Event type, Impact

2 modules

an index that follows the news, and a bank stress test

Yugh Juneja

VIT Bhopal University · github.com/yugh88/VIT_YughJuneja_Hackathon

PROBLEM & APPROACH

Risk arrives as text. Sentinel turns it into numbers a bank can act on.

THE PROBLEM



Market-moving news comes as text: wires, filings, central banks, social posts.



It arrives in dozens of languages, faster than any team can read.

❓ A risk desk must know fast: **who** is hit, **how badly**, **what it costs**.

OUR APPROACH

STEP 1

Collect

9 free sources in 65 languages, each with a recorded backup

→

STEP 2

Understand

Find the companies and countries; drop spam and copies

→

STEP 3

Score

Sentiment –1 to +1 · **Event** type · **Impact** 1–10, learned from real price moves

→

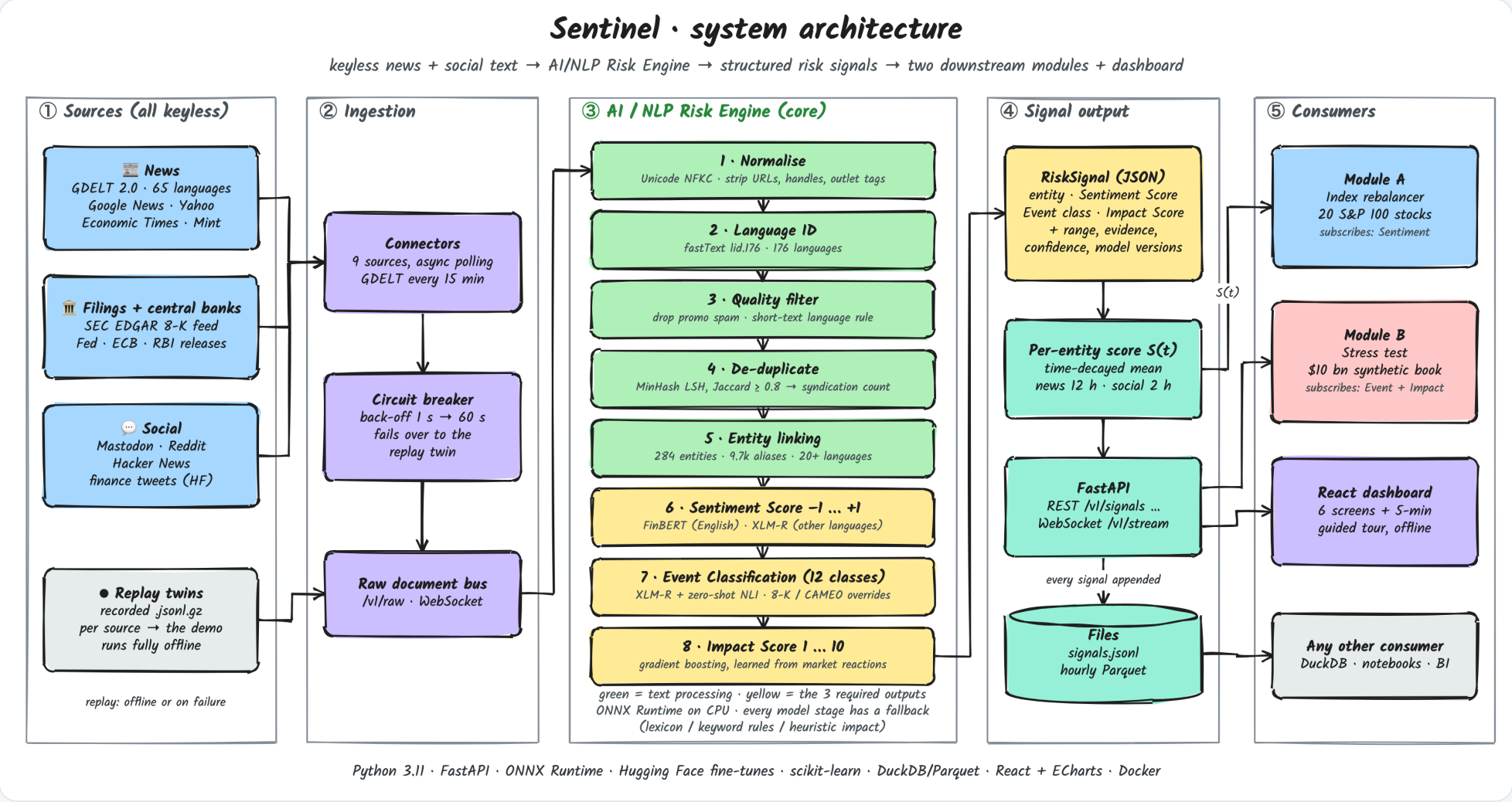
STEP 4

Act

A: re-weight an index
B: stress-test a $10 bn bank book

SYSTEM DESIGN

Five layers, from raw text to a decision



IMPLEMENTATION HIGHLIGHTS

Six choices that make it work

Finance-trained AI models

FinBERT for English; XLM-RoBERTa plus a multilingual NLI model for other languages.

Why: reads context, not keywords

Impact learned from prices

A gradient-boosting model trained on 74,964 news items and the price moves that followed.

Why: grounded in real reactions

Cheap filters first

64% of documents mention no tracked company and stop before the AI; duplicates merge.

Why: fast on a laptop CPU

Never goes dark

Every source has a recorded replay that takes over when it fails.

Why: the demo runs fully offline

Module A · Index rebalancer

20 S&P 100 stocks tilt towards good news, inside hard limits (15% per stock, 35% per sector).

Why: news-aware, risk-controlled

Module B · Stress test

A confirmed serious event re-prices 300 positions and runs 10,000 scenarios in 0.2 s.

Why: a capital number in seconds

Python 3.11 · FastAPI · ONNX Runtime · Hugging Face · scikit-learn · React + TypeScript · Docker · GitHub Actions

KEY RESULTS · MEASURED ON DATA THE MODELS NEVER SAW

Accurate, multilingual and fast

0.85

English sentiment score (F1)

word-list methods score 0.45

0.82

Event type score (F1)

12 event classes

92%

Credit events caught

new 2025–26 headlines

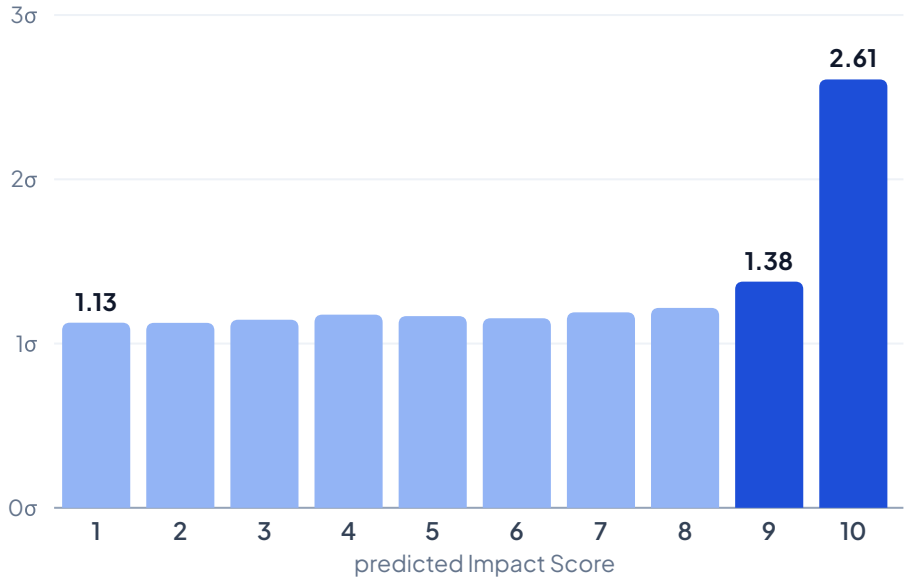
44 ms

Per document on a laptop CPU

64% of documents skip the AI entirely

Higher Impact Score → bigger real price moves

Average move after the news, by predicted score (2024+ news)



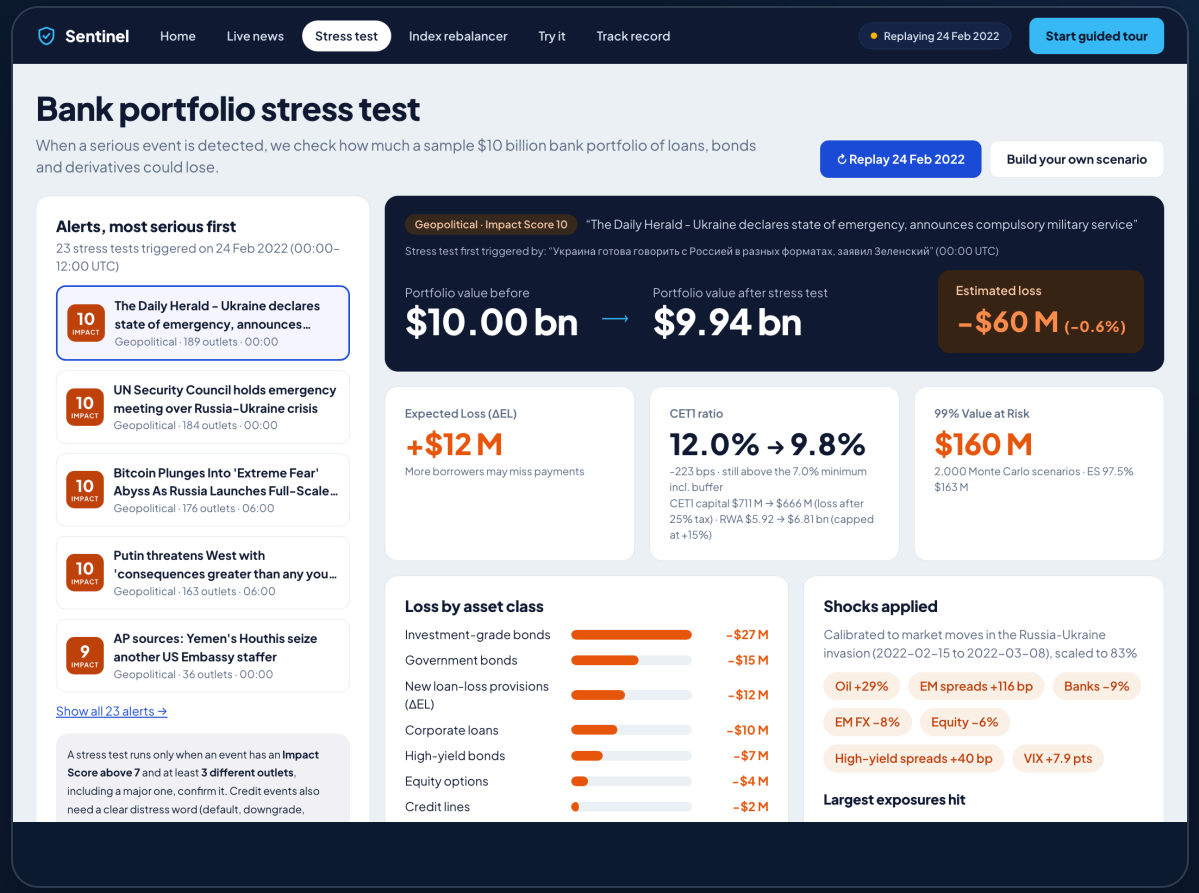
Impact 10 news moves prices 2.3× more than Impact 1.

FINAL ROUND Other languages **0.54 → 0.65** Credit precision **0.58 → 0.76** Response time **177 → 44 ms**

All results from held-out test sets; details in docs/model_card.md

DOMAIN IMPACT · THE MORNING OF 24 FEB 2022, REPLAYED WITH REAL NEWS

From headlines to a capital number, automatically



11,520 news articles

3,107 signals about a company or country

23 stress tests confirmed by several outlets

Top alert: Ukraine declares emergency

$10.00 bn → $9.94 bn

capital ratio 12.0% → 9.8% · 99% VaR $160 m

- **Early:** first alerts at 00:00 UTC, from Russian-language headlines
- **Actionable:** loss, capital and VaR, not just a red flag
- **Trustworthy:** one tweet cannot trigger a test; every score shows its evidence

Module B on a seeded synthetic $10 bn book · shocks calibrated on real episodes (Ukraine 2022, 2022 rate hikes, SVB 2023)

ASSUMPTIONS & NEXT STEPS

Built to grow into a production system

Assumptions

- Synthetic, seeded bank book; public data only
- Stress shocks scale with the Impact Score from real past episodes
- Recorded finance tweets stand in for X, which has no free API

Next steps

- Human-labelled native-language data to raise multilingual accuracy
- Connect real exposure feeds in place of the synthetic book
- GPU or INT8 workers for higher volume
- Intraday price labels for a sharper Impact Score

Scale **10 million documents a day** ≈ 7 CPU workers on average Stateless workers · one signal format · runs offline

Run it: make setup && make demo · github.com/yugh88/VIT_YughJuneja_Hackathon